Activation Functions

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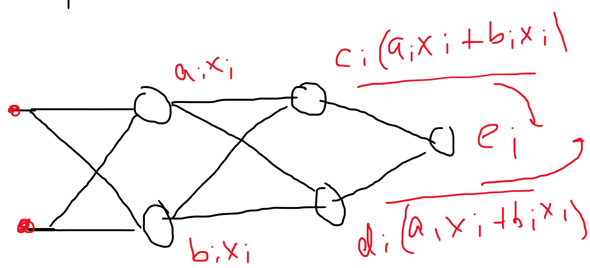
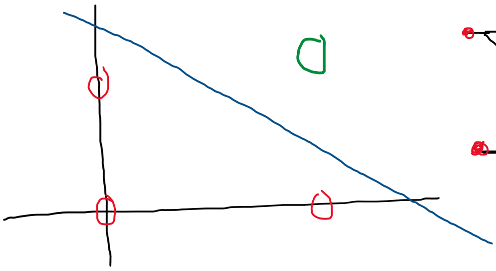
Activation Function:-

$$y = f(z) \quad \Big| \quad z = \sum_{i=0}^{n} w_i x_i$$

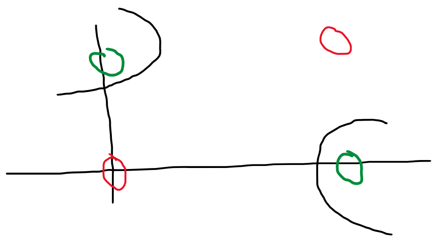
Def^n ← (non linearity introduce) (i) ✓



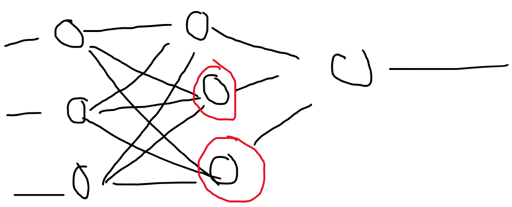
And gate



XOR gate



ii) Deciding which neurons fire



iii) Z transformation

Types of Activation Functions

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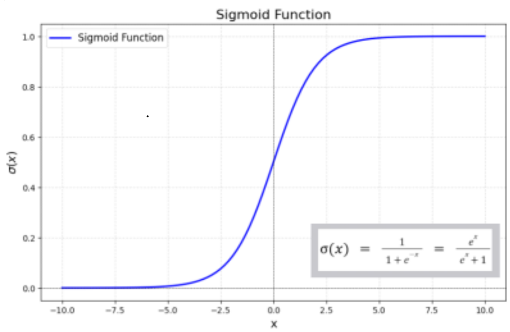
1. Sigmoid (classification) Logistic

$$\sigma(z) = \frac{1}{1+e^{-z}} = \begin{cases} 0 \\ 1 \end{cases}$$



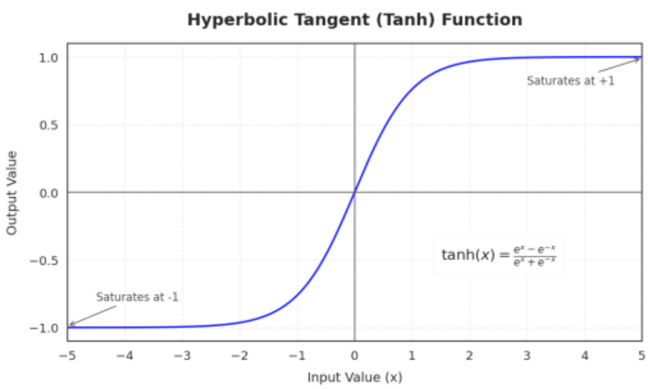
(0, 1) ✓

[0, 1] ✗

$$\boxed{\sigma'(z) = \sigma(z) \cdot (1 - \sigma(z))}$$

* Interpretable as Probability

2. Tanh (Hyperbolic Tangent) Function:-



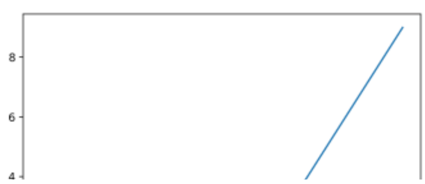
$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}} = \begin{cases} -1 \\ +1 \end{cases}$$

(-1, +1)

3. RELU:-

Rectified Linear Unit

$$f(z) = \max(0, z)$$



$$f(z) = \max(0, z)$$